Financial Fraud Detection: Anomaly Detection in Credit Card Transactions

Business Introduction

SecureGuard Financial Solutions is a company focused on providing advanced fraud detection systems for financial institutions. With the rapid growth of digital payments and online transactions, fraudulent activities such as unauthorized transactions and identity theft have become increasingly common. SecureGuard aims to analyze transaction data to identify suspicious patterns and help organizations prevent financial losses while maintaining customer trust.

Project Objective

The primary objective of this project is to analyze credit card transaction data and identify patterns that indicate fraudulent activity. By leveraging data analysis techniques, the project aims to support the development of a system that can detect fraud in real time and minimize risks for both customers and financial institutions.

Problem Statement

The goal is to design a system that can detect fraudulent credit card transactions by analyzing transaction data. The system should identify unusual spending behavior, abnormal transaction patterns, and anomalies that deviate from normal customer activity.

Dataset Description

The dataset used in this project contains detailed information about credit card transactions. Each record represents a transaction and includes attributes such as transaction amount, category, merchant details, customer demographics, location information, and a fraud indicator. The dataset enables analysis of both normal and fraudulent behavior patterns.

Excel Analysis

Statistical Summary (amt, city_pop)

A statistical summary was performed on the transaction amount and city population variables to understand their distribution and central tendencies.

city pop Statistical Summary	
Mean	88680.8429
Standard Error	482.940487
Median	2456
Mode	606
Standard Deviation	301210.102
Sample Variance	9.0728E+10
Kurtosis	37.5443115
Skewness	5.58789245
Range	2906677
Minimum	23
Maximum	2906700
Sum	3.4497E+10
Count	389002

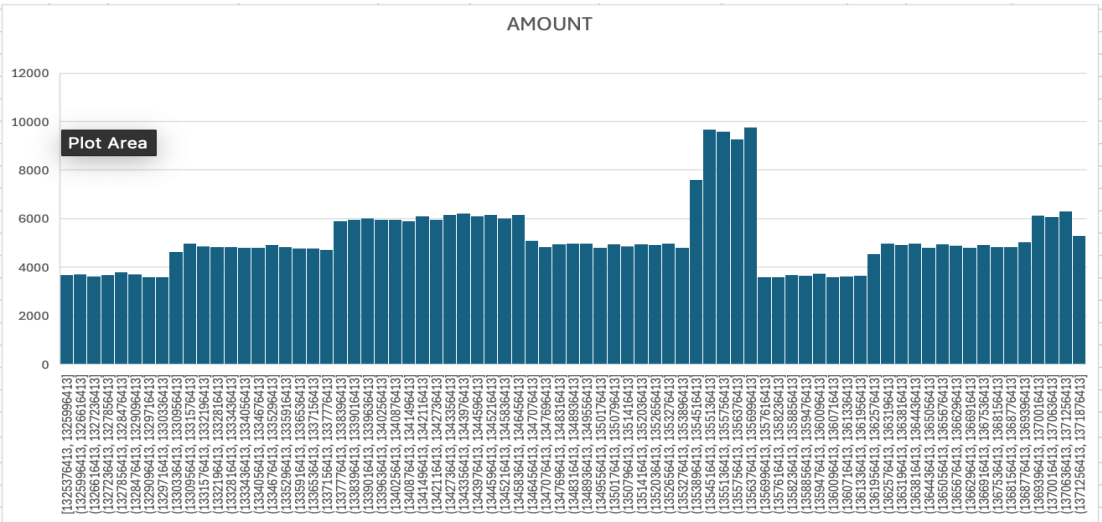
Amount Statistical Summary	
Mean	70.44214816
Standard Error	0.2600671
Median	47.57
Mode	1.04
Standard Deviation	162.2039152
Sample Variance	26310.11011
Kurtosis	3868.242869
Skewness	40.27383233
Range	27389.12
Minimum	1
Maximum	27390.12
Sum	27402136.52
Count	389002

Insights:

- The transaction amount shows a right-skewed distribution, indicating that most transactions are of lower value with a few high-value transactions.
- The presence of very high maximum values suggests potential outliers, which may indicate fraudulent activity.
- City population varies significantly, indicating that transactions occur across both small and large cities.

Histogram of Transaction Amount (amt)

A histogram was created to visualize the distribution of transaction amounts.



Insights:

- The distribution is heavily right-skewed, with most transactions concentrated in lower ranges.
- A long tail is observed, representing high-value transactions.
- These extreme values are important as they may correspond to suspicious or fraudulent transactions.

Fraud Count by Gender and Category

A pivot table was used to analyze the number of fraudulent transactions across different genders and transaction categories.

Sum of is_fraud	gender		
category	F	M	Grand Total
entertainment	46	22	68
food_dining	35	7	42
gas_transport	88	105	193
grocery_net	17	23	40
grocery_pos	229	289	518
health_fitness	33	6	39
home	60	5	65
kids_pets	57	17	74
misc_net	120	167	287
misc_pos	51	26	77
personal_care	58	18	76
shopping_net	188	290	478
shopping_pos	137	116	253
travel	22	20	42
Grand Total	**1141**	**1111**	**2252**

Insights:

- Certain transaction categories show higher fraud occurrences, indicating they are more vulnerable to fraudulent activities.
- Fraud distribution across gender is relatively balanced, suggesting that fraud is not strongly dependent on gender.
- Category-based analysis is more useful for identifying fraud patterns.

Top 3 States by Number of Transactions

A pivot table was used to identify the states with the highest number of transactions.

Top 3 states by transaction	
states	**Count of transaction number**
TX	28482
NY	25034
PA	23911
Grand Total	**77427**

Insights:

- A few states contribute a large portion of total transactions.
- High transaction volume areas represent higher exposure to potential fraud.
- However, high transaction count does not necessarily imply high fraud rate.

Correlation between Transaction Amount and City Population

Correlation analysis was performed between transaction amount and city population.

Correlation between AMOUNT and CITY POP
0.007271

Insights:

- The correlation between transaction amount and city population is weak to moderate.
- This indicates that while larger cities may have slightly higher transaction values, the relationship is not strong.
- Transaction behavior is influenced by multiple factors beyond city size.

Fraud Transactions by Category

A pivot table was created to analyze fraud distribution across categories.

Insights:

category	Sum of is_fraud
entertainment	68
food_dining	42
gas_transport	193
grocery_net	40
grocery_pos	518
health_fitness	39
home	65
kids_pets	74
misc_net	287
misc_pos	77
personal_care	76
shopping_net	478
shopping_pos	253
travel	42
Grand Total	**2252**

- Certain categories have a significantly higher number of fraudulent transactions.
- These categories can be considered high-risk and should be monitored more closely.
- Category-based patterns are useful for building fraud detection rules.

Average Transaction Amount by Job

A pivot table was used to calculate the average transaction amount across different job roles.

JOBS	Average of amt
Air traffic controller	503.748
Broadcast journalist	1011.325
Careers adviser	531.466
Dancer	586.45
Engineer, site	638.9125
Industrial buyer	654.49
Personnel officer	638.5766667
Sales promotion account executive	529.6485714
Ship broker	972.11
Veterinary surgeon	655.05
Grand Total	**610.2688889**

Insights:

- Spending behavior varies across different professions.
- These are Top 10 jobs average amount.
- Some job categories show higher average transaction amounts.
- This variation can help in identifying abnormal transactions when compared to expected spending patterns.